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Lymphatic filariasis in Uganda: baseline investigations in Lira, Soroti and Katakwi districts

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Abstract

Baseline epidemiological investigations on lymphatic filariasis were conducted for the first time in Uganda in 3 communities in the districts of Lira (Alebtong area), Soroti (Lwala area) and Katakwi (Obalanga area), located to the north of Lake Kyoga at an altitude of 1000–1100 m above sea level. Individuals from the communities were examined, in April–August 1998, for *Wuchereria bancrofti* specific circulating antigen (by ICT card test), microfilaraemia (by counting chamber and stained blood-smear techniques) and chronic clinical manifestations of lymphatic filariasis. Endophilic mosquitoes were sampled and dissected for filarial larvae. Prevalences of circulating filarial antigen positivity were 29%, 18% and 30% in the Alebtong, Lwala and Obalanga communities, respectively. Microfilaria (mf) prevalences were 18%, 9% and 21%, and geometric mean mf intensities among mf-positive individuals were 306, 171 and 402 mf/mL blood, in the same communities. Examination of stained blood smears revealed mf of both *W. bancrofti* and *Mansonella perstans*, but more than 80% of mf-positive individuals harboured the first of these parasites. Prevalences of hydrocoele in adult (≥ 20 years) males were 28%, 7% and 17%, and prevalences of limb elephantiasis in adults were 9%, 4% and 4%, in the Alebtong, Lwala and Obalanga communities, respectively. *Anopheles gambiae s.l.* (mainly *An. gambiae s.s.*) and *An. funestus* were common in all 3 communities, and showed *W. bancrofti* infectivity rates of 1·1–1·7% and 1·3–2·9%, respectively. It is concluded that lymphatic filariasis is highly endemic in these high-altitude areas of Uganda, with *An. gambiae s.l.* and *An. funestus* being the main vectors.

Keywords: lymphatic filariasis, hydrocoele, lymphoedema, microfilariae, circulating antigens, *Wuchereria bancrofti*, *Mansonella perstans*, vector mosquitoes, Uganda

Introduction

Lymphatic filariasis, resulting from infection with the mosquito-borne parasitic nematode *Wuchereria bancrofti*, is widely distributed in tropical and subtropical areas of the world, where it results in considerable suffering and debilitating clinical disease (MCMAHON & SIMONSEN, 1996). Recent estimates indicate that more than 50 million people in sub-Saharan Africa are affected (MICHAEL & BUNDY, 1997), thus imposing more than one-third of the global burden of lymphatic filariasis to this region. Major international efforts are now being mobilized to promote and support lymphatic filariasis control activities in endemic areas (WHO, 1999).

In Uganda, most reports on lymphatic filariasis have been based on clinical cases seen in hospitals. As such it has been reported from many parts of the country, although little has been documented in the literature. HAWKING (1943) was the first to draw attention of the medical and scientific communities to the problem, by summarizing information available from local health reports in the then Uganda Protectorate. Later, BURKITT (1951) reported on the prevalence of hydrocoele, and on hydrocoelectomies, in adult males in what is now Lira District, and SPENCER (1962) conducted a clinical study of lymphatic filariasis cases admitted at Lira District Hospital. These observations, together with later updates by HAWKING (1957, 1977) provide the only published documentation so far available on lymphatic filariasis in Uganda. Thus, to date, its geographical distribution, infection characteristics, clinical spectrum, transmission dynamics and socio-economic impact are largely unknown.

Reports from Ministry of Health field staff in different parts of the country suggest that lymphatic filariasis due to *W. bancrofti* is still common and constitutes a serious health problem in many parts of Uganda. The purpose of the present study was to carry out baseline epidemiological investigations on lymphatic filariasis, including surveys to identify the important vectors, in a part of Uganda from where it has repeatedly been reported to be of major health concern.

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Materials and Methods

Study area

The surveys were carried out during April–August 1998 in the districts of Lira, Soroti and Katakwi, located to the north of Lake Kyoga in Uganda (Fig. 1). In each district, 4–5 adjacent villages were selected in order to get a study population of at least 1000 individuals per district. Four villages (Eyama, Anang, Ipale, Amuria) were in the Alebtong area of Lira District; 4 (Acetangorom, Awimon, Oculoi, Olemai) in the Lwala area of Soroti District; and 5 (Aberan, Abota, Agaya, Akora, Amenangura) in the Obalanga area of Katakwi District.

The 3 study areas are quite similar in terms of physical features, vegetation, human settlement and economic activities. The altitude is 1000–1100 m above sea level. Mean minimum and maximum temperatures are 15·0–20·0°C (June/July) and 27·0–33·0°C (February/March), respectively. The annual rainfall is 1000–1500 mm, with a main rainy season from April to June. The countryside is generally flat, and seasonal swamps and streams are widespread. The predominant natural vegetation is that of wet and dry savannah. The areas have low population density and settlements within villages are widely scattered. Individual homesteads usually comprise 2–3 houses, the majority of which have grass-thatched roofs, and mud and wattle walls. Roads (often tracks or paths) to and within the villages are poor. There are 3 major tribal groups, namely the Lango (predominant in Alebtong), the Kumam (predominant in Lwala) and the Ateso (predominant in Obalanga). The main occupation is subsistence farming, and common cultivated crops are cotton, maize, beans, groundnuts, sesame, cassava, sweet potatoes and millet. Cattle-keeping used to be the mainstay of the areas, but years of rustling have considerably depleted the cattle population. Drug treatment for lymphatic filariasis has generally not been available, but hydrocoelectomies are commonly performed in the Lwala area, at Lwala Mission Hospital. Onchocerciasis is not transmitted in the areas.

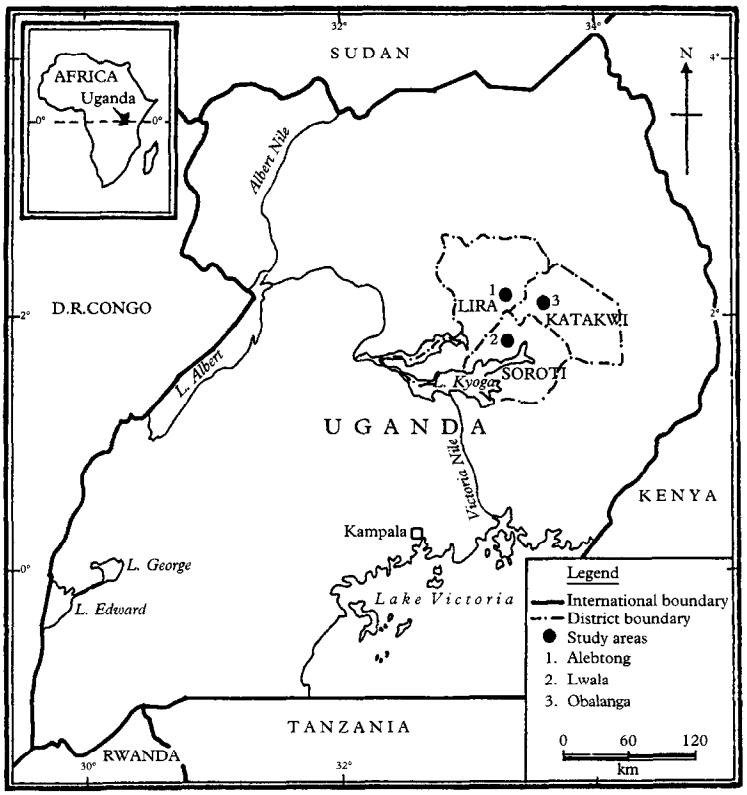


Fig. 1. Map of Uganda showing the 3 study areas: Alebtong (Lira District), Lwala (Soroti District) and Obalanga (Katakwi District).

Study design

After the subject of activities had been introduced to the local authorities, community leaders and villagers, and their permission to carry out the surveys had been obtained, all residents of the selected villages were registered. Individuals were then requested to come to a central point of the village during daytime for clinical assessment and for investigation for circulating antigenaemia, and to return on the same evening between 21:00 and 02:00 for examination for nocturnal microfilaraemia. Houses in each community were furthermore selected for spray catches and light-trap catches, to identify common domestic vector mosquitoes. The study was approved by the Ministry of Health, Uganda. Oral informed consent to participate was obtained from individuals aged $\geqslant 15$ years, and from the parents of younger individuals.

Clinical examinations

Individuals aged $\geqslant 1$ year, who had lived in the villages for at least 1 year, were clinically examined for obvious chronic manifestations of lymphatic filariasis. Thus, males and females were examined for limb lymphoedema/elephantiasis. Males were examined for hydrocoele and other scrotal involvement. Signs of current attacks of adenolymphangitis (ADL) and of scarification were also recorded. Chronic signs were graded as described by ESTAMBALE *et al.* (1994) but, for simplicity in this presentation, these are reported as true hydrocoele ($\geqslant 6$ cm, with fluid accumulation), limb elephantiasis (includes both lymphoedema and true elephantiasis) and scrotal elephantiasis, only.

Examinations for circulating filarial antigens

Immediately after clinical examination, those who agreed to have a blood sample taken were examined for circulating filarial antigens (CFA) by use of the ICT whole blood card test for *W. bancrofti* antigens (Amrad ICT, Australia). In brief, 100 µL of fingerprick blood was drawn from each donor, using a heparinized capillary tube, and the blood was transferred to an ICT card. The card was closed, and the result was read after 2 min.

Examinations for microfilariae

Individuals who had been clinically examined were requested to return for parasitological examination in the evening between 21:00 and 02:00. Owing to the long walking distances within the scattered settlements and the late hours many were unable to come. Indeed, some of those who came had to spend the night where the examination was held. From each donor, 100 µL of fingerprick blood was collected using a heparinized capillary tube. The blood was preserved in specimen bottles with 0·9 mL of 3% acetic acid. Samples were later examined for microfilariae (mf) under a microscope using the counting-chamber technique (MCMAHON *et al.*, 1979). The number of mf per 100 µL of blood was counted and the mf intensity per mL of blood was calculated.

For the purpose of specific identification of mf species, blood smears were prepared at the same time. Only individuals from whom additional blood could easily be obtained from the original fingerprick puncture were included, and the coverage was therefore lower than for the counting-chamber examinations. Blood smears were dried, dehaemoglobinized, fixed, stained with Giemsa stain and examined under a microscope. mf were identified to species based on morphological criteria (WHO, 1987).

Entomological surveys

Mosquitoes were sampled using pyrethrum-spray catch (PSC) and light-trap (LT) methods (SERVICE, 1993). CDC light-traps, hung just outside an occupied bednet, were allowed to run from 18:00 in the evening until they were emptied in the morning at 06:00. Spray catches were performed in the morning, between 08:00 and 11:00. Adult female mosquitoes sampled were identified to species on morphological criteria. A subsample of *An. gambiae s.l.* and *An. funestus* was dissected

in distilled water. Dissections of the heads, thoraces and abdomens were examined fresh for motile infective (L3) and other (L1 and L2) filarial larval stages. As a control, dissections were thereafter dried, fixed, stained and examined as described by McMAHON *et al.* (1981). Filarial larvae were then identified based on morphological criteria (SASA, 1976; WHO, 1987). Another sub-sample of *An. gambiae s.l.* was preserved in 80% iso-propanol for sibling-species identification to either *An. gambiae s.s.* or *An. arabiensis* by polymerase chain reaction (PCR; SCOTT *et al.*, 1993).

Data analysis

W. bancrofti has nocturnal microfilarial periodicity in Africa. In order to adjust for the effect of sampling time on mf intensities, observed intensities were multiplied with a time-specific factor as previously described (SIMONSEN *et al.*, 1997). mf geometric mean intensities (GMI) were calculated as antilog $[(\sum\log(x+1))/n]-1$, with x being the number of mf/mL in mf-positive individuals and n the number of mf-positive individuals examined. Differences in prevalences were analysed by χ^2 test, and differences in means were analysed by t-test or 1-way analysis of variance, as appropriate. P-values $< 0{\cdot}05$ were considered statistically significant.

Results

The registered populations in the Alebtong, Lwala and Obalanga study communities were 1804, 2164 and 1602, respectively. Of these, 1180 (65·4%), 1124 (51·9%) and 1244 (77·7%) individuals participated in the clinical examination. More females than males were examined in all 3 communities (Table 1).

Clinical manifestations

The most common filariasis-related clinical manifestation in all communities was hydrocoele. Among men aged ≥ 20 years, prevalences of hydrocoele were 28·7%, 7·0% and 17·3%, in Alebtong, Lwala and Obalanga, respectively (Table 1). The youngest victims of hydrocoele were aged 16, 23 and 13 years in these communities, and prevalences increased gradually by age to reach 45·8%, 10·5% and 22·9% in men aged ≥ 50 years. Scrotal elephantiasis was seen in 17, 3 and 2 males (age range 25–85 years) in Alebtong, Lwala and Obalanga, respectively (8·1%, 1·5% and 0·9% of men aged ≥ 20 years).

Overall prevalences of limb elephantiasis in individuals aged ≥ 20 years were 9·7%, 3·9% and 3·7% in Alebtong, Lwala and Obalanga, respectively (Table 1). The youngest individuals with limb elephantiasis were aged 18, 18 and 11 years, and prevalences increased gradually by age to reach 23·0%, 8·0% and 9·2% in individuals aged ≥ 50 years. Most cases were in the legs, and 52·2% of these were bilateral. Only 2 cases of arm elephantiasis were seen, both of which were unilateral (female in Alebtong, 62 years; male in Obalanga, 40 years). In all communities limb elephantiasis was more common in males than in females. Thus, in individuals aged ≥ 20 years the ratios of male to female prevalence were 5·2, 2·0 and 1·9 in Alebtong, Lwala and Obalanga, respectively.

Signs of ongoing ADL attacks were seen in 6, 3 and 3 individuals in Alebtong, Lwala and Obalanga, respectively. Scarification of the affected limb was commonly observed in individuals with elephantiasis (e.g., in all 11 females with elephantiasis in Alebtong). It should also be pointed out that many individuals had more than one chronic manifestation. In Lira, for example, 10 males had a combination of scrotal elephantiasis, limb elephantiasis and hydrocoele, and 7 males had both scrotal elephantiasis and hydrocoele.

Circulating antigens

Overall prevalences of CFA positivity were 29·1%, 18·3% and 30·1% in Alebtong, Lwala and Obalanga, respectively. The youngest CFA-positive individuals were aged 1, 4 and 1 years, and prevalences of CFA positivity gradually increased by age to 47·1%, 35·0% and 39·6% in individuals aged ≥ 50 years (Fig. 2). Among adults (≥ 20 years) in all 3 communities, prevalences of CFA positivity were higher in men than in women (Fig. 3A), and this difference was statistically significant for Alebtong and Lwala ($P = 0{\cdot}01$ and $< 0{\cdot}001$, respectively).

Table 1. Population characteristics, chronic clinical manifestations related to *W. bancrofti* infection, circulating filarial antigenaemia and microfilaraemia in the three study communities (Uganda, 1998)

	Alebtong	Lwala	Obalanga
Registered population (aged ≥ 1 year)	1804	2164	1602
Clinically examined population	1180	1124	1244
Male : female ratio	1 : 1·15	1 : 1·35	1 : 1·13
Mean age (years)	22·1	23·5	20·7
Hydrocoele in males ≥ 20 years			
No. examined	209	201	231
No. positive (%)	60 (28·7)	14 (7·0)	40 (17·3)
Limb elephantiasis in individuals ≥ 20 years			
No. examined	495	518	534
No. positive (%)	48 (9·7)	20 (3·9)	20 (3·7)
Circulating filarial antigens (CFA)			
No. examined	890	896	900
No. positive (%)	259 (29·1)	164 (18·3)	271 (30·1)
Microfilariae (counting-chamber technique)			
No. examined	739	572	799
No. positive (%)	136 (18·4)	51 (8·9)	165 (20·7)
GMI on mf positives (mf/mL)	306	171	402
Microfilariae (stained blood smear)			
No. examined	322	422	513
No. positive for mf* (%)	87 (27·0)	45 (10·7)	128 (25·0)
No. positive for *W. bancrofti* mf (%)	82 (25·5)	41 (9·7)	115 (22·4)
No. positive for *M. perstans* mf (%)	16 (5·0)	10 (2·4)	24 (4·7)

*11, 6 and 11 slides from Alebtong, Lwala and Obalanga, respectively, had mf of both species.
GMI, geometric mean intensity.

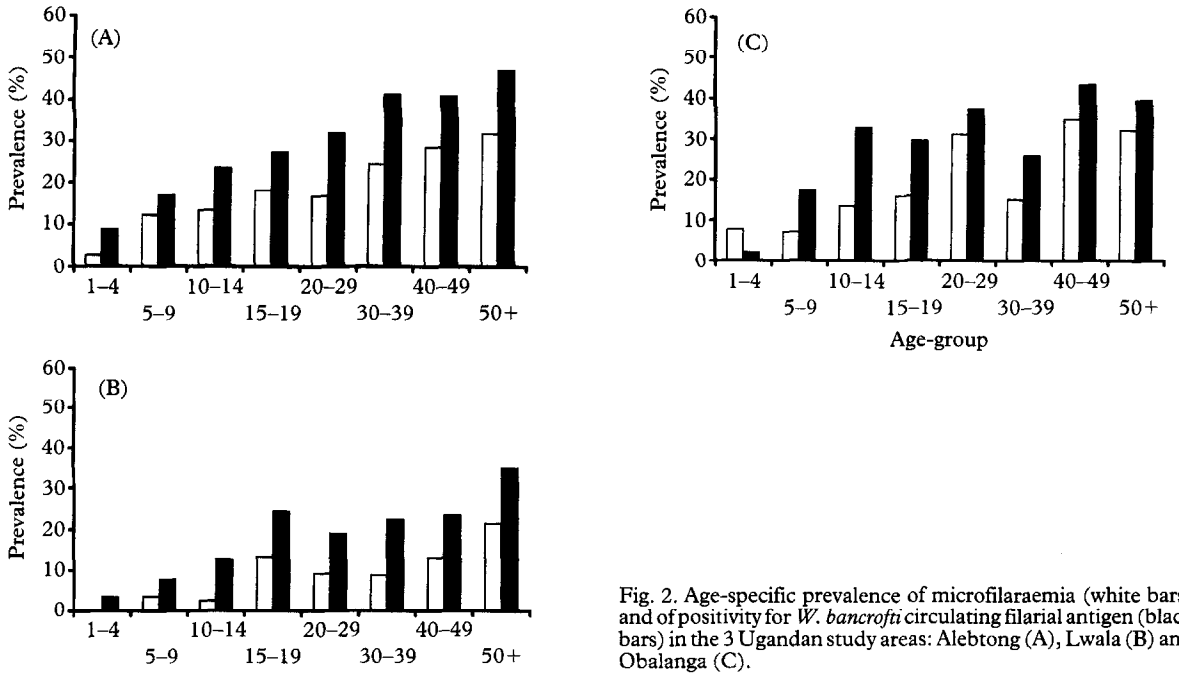


Fig. 2. Age-specific prevalence of microfilaraemia (white bars) and of positivity for *W. bancrofti* circulating filarial antigen (black bars) in the 3 Ugandan study areas: Alebtong (A), Lwala (B) and Obalanga (C).

Microfilaraemia

Based on counting-chamber technique assessment, overall mf prevalences were 18·4%, 8·9% and 20·7% in Alebtong, Lwala and Obalanga, respectively. The youngest mf-positive individuals were aged 4, 8 and 1 years, and mf prevalences increased by age to 31·8%, 21·5% and 32·2% in individuals aged ≥ 50 years (Fig. 2). Among adults (≥ 20 years) in all 3 communities, the mf prevalence was higher in men than in women (Fig. 3B), and this difference was statistically significant for Lwala and Obalanga ($P < 0{\cdot}001$ and $= 0{\cdot}04$, respectively). For all villages combined, 20·3% and 21·8% of adults with hydrocoele and elephantiasis, respectively, were mf positive, and 28·7% and 28·9% were CFA positive.

Among mf-positive individuals, mf intensity ranges were 10–9641, 12–2922 and 10–13356 mf/mL in Alebtong, Lwala and Obalanga, respectively. The mf GMI among mf-positive individuals was highest in Obalanga (402 mf/mL), followed by Alebtong (306 mf/mL) and Lwala (171 mf/mL). mf GMIs did not differ significantly between age-groups (5–9, 10–14, 15–19, 20–29, 30–39, 40–49 and 50+ years) in any of the communities ($P > 0{\cdot}05$ for all tests). The mf GMI was significantly higher in males than in females in Obalanga ($P = 0{\cdot}01$), whereas no significant gender differences were detected in Alebtong or Lwala.

Blood smears for identification of microfilarial species indicated that 27·0%, 10·7% and 25·0% of these were positive for mf in Alebtong, Lwala and Obalanga, respectively (Table 1). Examination revealed that mf in the smears belonged to 2 species, namely *W. bancrofti* and *M. perstans*. However, *W. bancrofti* were by far the more common, being present in 83·7%, 80·4% and 82·7% of mf-positive slides in the 3 communities.

Among individuals who were positive for mf of *W. bancrofti* in blood smears and who were also examined by card test for CFA, all were CFA positive (64, 31 and 93 individuals in Alebtong, Lwala and Obalanga, respectively). Among individuals found mf positive by the counting-chamber technique and who were also examined by card test for CFA (105, 42 and 148, respectively), 100·0%, 97·6% and 95·3% were also CFA positive.

Entomological results

Table 2 summarizes data from the spray and light-trap catches. In Alebtong and Obalanga *An. gambiae s.l.* and *An. funestus* were the predominant species, whereas culicine mosquitoes dominated in Lwala. Small samples of *An. gambiae s.l.* females were examined for sibling-species identity by PCR (50, 30 and 50 from Alebtong, Lwala and Obalanga, respectively). Of these, 115 (88·5%) were successfully identified, and 98·3% (113) were *An. gambiae s.s.* Two mosquitoes from Obalanga were *An. arabiensis*. The majority of culicine mosquitoes collected in all 3 areas were *Mansonia africana*, and no *Culex quinquefasciatus* was seen.

All 3 areas had comparatively high mosquito densities. The numbers of anopheline mosquitoes caught by light

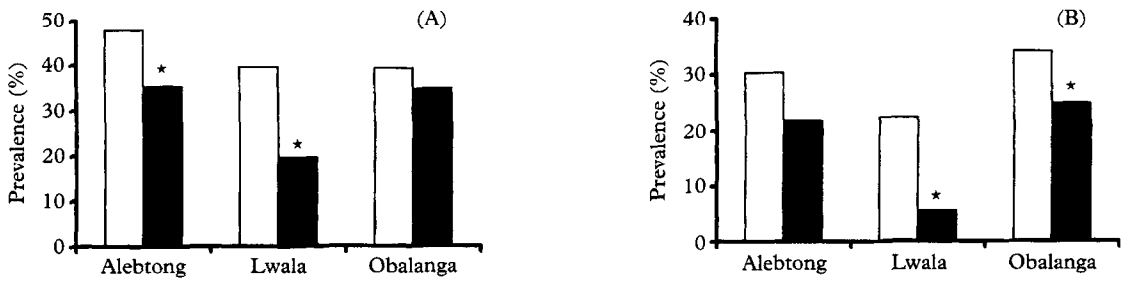


Fig. 3. Prevalence of positivity for *W. bancrofti* circulating filarial antigen (A) and for microfilaraemia (B) among males (white bars) and females (black bars) in the 3 Ugandan study communities. Statistically significant differences in prevalence between males and females within the community are indicated with an asterisk over the female bar.

Table 2. Number of mosquitoes caught in light traps and by pyrethrum-spray catch (PSC), and their infection and infectivity rates, in the three study communities

	Alebtong	Lwala	Obalanga
No. of light-trap catches	29	50	31
No. of PSCs	23	37	27
Anopheles gambiae s.l.			
No. in light trap (mean no./catch)	871 (30·0)	216 (4·3)	2392 (77·2)
No. in PSC (mean no./catch)	839 (36·5)	328 (8·9)	1793 (66·4)
No. dissected	451	242	358
No. infected (%)	14 (3·1)	16 (6·6)	10 (2·8)
No. infective (%)	7 (1·6)	4 (1·7)	4 (1·1)
Anopheles funestus			
No. in light trap (mean no./catch)	397 (13·7)	392 (7·8)	2526 (81·5)
No. in PSC (mean no./catch)	240 (10·4)	355 (9·6)	1779 (65·9)
No. dissected	174	155	265
No. infected (%)	11 (6·3)	10 (6·5)	12 (4·5)
No. infective (%)	5 (2·9)	2 (1·3)	4 (1·5)
Other anophelines			
No. in light trap* (mean no./catch)	31 (1·1)	21 (0·4)	24 (0·8)
Culicines			
No. in light trap** (mean no./catch)			
Mansonia africana	86 (3·0)	2097 (41·9)	1020 (32·9)
Other culicines	193 (6·7)	566 (11·3)	70 (2·3)
No. in PSC** (mean no./catch)	9 (0·4)	50 (1·4)	40 (1·5)

Mean catch is calculated as the arithmetic mean.
*Mainly *An. coustani*, but *An. squamosus*, *An. nili* and *An. pharaoensis* were also seen. Only *An. gambiae* and *An. funestus* were seen in PSC.
**None of these was *Culex quinquefasciatus*.

traps and by spray catch were comparable, whereas substantially fewer culicine mosquitoes were collected by spray catch than by light-trap catch. Obalanga had the highest mosquito density (195 per light-trap catch; 81·9% of these were anophelines). Alebtong had fewer mosquitoes than Obalanga (54·4 per light-trap catch; 82·3% of these were anophelines). Lwala had about the same mosquito density as Alebtong (65·8 per light-trap catch; 19·1% of these were anophelines), but culicine mosquitoes dominated.

Infective larvae of *W. bancrofti* were found in both *An. gambiae s.l.* and *An. funestus* from all 3 communities (Table 2). Thus, infectivity rates were 1·6, 1·7 and 1·1% in *An. gambiae* and 2·9, 1·3 and 1·5% in *An. funestus* in Alebtong, Lwala and Obalanga, respectively.

Discussion

We document here for the first time an epidemiological baseline study on lymphatic filariasis in Uganda. Previous reports had indicated that lymphatic filariasis occurred in the country (HAWKING, 1943, 1957, 1977; BURKITT, 1951; SPENCER, 1962) but little attention had been paid to its epidemiology, transmission or public-health importance. The present surveys showed that lymphatic filariasis is highly endemic and a major health problem in the examined communities.

The surveyed areas had poor infrastructure, and settlements were widely scattered. This was an impediment to survey activities, especially night work, since the villagers had to walk long distances. Other untoward events affected the number of people attending examinations in one or more of the communities after registration. Thus, a tax operation to net defaulters coincided with the survey and some families kept away for fear that they would be arrested; members of a certain religious sect refused to be examined; some individuals believed the exercise was an HIV surveillance and were scared away; some male youth were recruited to work in tea estates elsewhere; and some individuals refused since they felt that their participation would give political recognition to a certain association whose members were collaborating with the survey team. Despite this, reasonable survey coverages were attained, and there is no indication that certain groups of examined individuals are misrepresented compared to the overall population.

In all 3 communities, hydrocoele was the most common chronic manifestation of lymphatic filariasis, and it occurred with high prevalence in the adult males. BURKITT (1951) and SPENCER (1962) had previously drawn attention to the high frequency of hydrocoele among male patients at Lira Hospital. In fact, it was here that BURKITT (1951) developed a surgical technique for hydrocoelectomy, involving eversion of the tunica vaginalis to prevent recurrence of the hydrocoele, which is now widely used in filariasis-endemic areas throughout the world. The close proximity of the Lwala study villages to Lwala Hospital probably to some extent contributed to the lower prevalence of hydrocoele recorded in this area. Records indicated that Lwala Hospital devotes 10–15% of its theatre time to hydrocoelectomies. In Obalanga, the scourge of hydrocoele has been instrumental in the formation of a powerful public movement, the Obalanga Hydrocoele Development Association, which aims at raising local awareness about filariasis and other issues of under-development, and methods for their solution.

Limb elephantiasis was the second most common chronic manifestation of lymphatic filariasis. Most cases were in the legs, and many were badly affected by secondary infections. According to local beliefs, elephantiasis is due to stepping on or over hidden 'magic', and only few victims would normally seek redress from modern medical practitioners. Rather, they would prefer self-treatment or consult traditional doctors. Many individuals with limb elephantiasis showed signs of multiple scars on the affected limbs, and scarification appeared to be a common traditional remedy to relieve the oedema as previously also reported from Ghana (DUNYO *et al.*, 1997). Scrotal elephantiasis was especially common in Alebtong. The reason for the high prevalence in this area is unclear, but SPENCER (1962)

similarly reported that scrotal elephantiasis and lymph scrotums were frequently seen in Lira Hospital.

Hydrocoele and elephantiasis often had considerable disabling effects on those affected, and were of major concern to the communities. The overall prevalences of these manifestations, and their patterns of distribution in the study communities, compared well with those reported from highly endemic foci in other parts of Africa (MCMAHON *et al.*, 1981; ESTAMBALE *et al.*, 1994; MEYROWITSCH *et al.*, 1995; SIMONSEN *et al.*, 1995; DUNYO *et al.*, 1996; DZODZOMENYO *et al.*, 1999). Only few episodes of acute ADL were recorded. However, ADL attacks usually last only a few days, and longitudinal surveys are necessary to document their frequency and health impact.

Examination of blood specimens for CFA and mf confirmed the presence of *W. bancrofti* infection in the examined populations. The stained blood smears furthermore showed mf of *M. perstans*, but with much lower prevalence. In fact, more than 80% of mf-positive individuals had mf of *W. bancrofti*. *M. perstans* is generally considered to be non-pathogenic, and has previously been reported from many parts of Uganda (HAWKING, 1943; FISCHER *et al.*, 1996). According to the manufacturers of the CFA cards, the test does not cross-react with circulating antigen from *M. perstans*. In accordance with this, 14 of 22 individuals in the present study who had only *M. perstans* mf in blood smears were CFA negative. In contrast, all *W. bancrofti* blood smear-positive individuals were CFA positive (100% sensitivity). It is possible, however, that some of the mf observed in the counting chamber may have been *M. perstans*, and counting-chamber results, including mf intensities, therefore have to be interpreted with some caution.

In general, and in accordance with findings from other studies (LAMMIE *et al.*, 1994; SIMONSEN *et al.*, 1996), CFA prevalences were higher than mf prevalences both at community level and within age-groups, probably because the CFA originate from all stages of the parasite and not only from the mf. The level of CFA positivity might therefore be a better indicator of the burden of infection and morbidity in the population than microfilaraemia. Age patterns of microfilaraemia and circulating antigenaemia followed those previously described from other endemic foci in Africa (ESTAMBALE *et al.*, 1994; MEYROWITSCH *et al.*, 1995; SIMONSEN *et al.*, 1995; DUNYO *et al.*, 1996; DZODZOMENYO *et al.*, 1999). In all 3 communities, prevalences of mf and CFA positivity were higher in males than in females. Prevalences of limb elephantiasis were also considerably higher among males than females. Gender differences in infection and disease patterns in lymphatic filariasis have been reported previously. The cause is unclear and may involve both physiological and behavioural factors (ALEXANDER & GRENFELL, 1999). Most individuals with chronic manifestations were negative for infection (mf and/or CFA), probably because manifestations develop over long periods.

Mosquito densities in the study areas were high. Light-trap catches revealed that *An. gambiae s.l.*, *An. funestus* and culicines, especially *Mansonia africana*, were the common endophagic mosquitoes. The high numbers of anophelines compared to culicines in the spray catches indicate that the former rest indoors whereas the latter, after taking a blood meal, rest outdoors. *An. gambiae s.s.* was the predominant sibling species of the *An. gambiae* complex in all the 3 areas.

An. gambiae s.l. and *An. funestus* are the common vectors of lymphatic filariasis throughout Africa, and these were caught in high numbers in the study communities. No specimens of *Culex quinquefasciatus*, which are common vectors of filariasis in coastal East Africa (BØGH *et al.*, 1998), were caught. *M. africana* is generally not considered a vector of filariasis. *An. gambiae s.l.* and *An. funestus* were dissected for filarial infection, and infected and infective specimens of both species were found in all 3 communities, and both in light-traps and in spray catches. The study therefore, for the first time, incriminated *An. gambiae s.l.* and *An. funestus* as principal vectors of lymphatic filariasis in rural Uganda. The 2 species appear to be of equal importance as vectors in the study communities. The density of anopheline mosquitoes in the 3 communities related well to the prevalence of filarial infection in humans. Thus, Obalanga with the highest anopheline density also had the highest prevalence of mf and CFA, and Lwala with the lowest anopheline density had the lowest prevalence of mf and CFA.

So far, most surveys for lymphatic filariasis in Africa have been carried out in lowland coastal areas, and it has been a common belief that filariasis occurs with highest prevalence and intensity in these hot and humid environments. The present study clearly documents that filariasis also occurs with high endemicity on the Uganda highland plains at about 1000–1100 m above sea level, with patterns of infection and disease comparable to those seen in coastal Africa. In spite of the altitude, temperatures are relatively high, and areas are densely covered by swamps and streams that support intense breeding of mosquito vectors. LINDSAY & THOMAS (2000), who analysed the population risk for lymphatic filariasis in Africa in relation to climatic conditions supporting vector breeding and parasite transmission, concluded that climates in large parts of Uganda, including the surveyed areas, are favourable for lymphatic filariasis transmission. Further studies will show the extent to which other parts of Uganda are also endemic for lymphatic filariasis.

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Announcement

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